A proof of the conjectured recurrence for OEIS A359643

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Abstract

OEIS A359643 carries a conjectured linear recurrence with polynomial coefficients, of order 5. The entry posts no generating function; it defines the sequence by an extraction, $a(n) = [x^n]\, f(x)g(x)^n$. We derive a generating function from that definition and prove the conjecture from it. Reading the extraction as a contour integral and summing the geometric series turns $\sum_n a(n)t^n$ into a single integral whose only pole inside a small circle is the branch of $x = t\,g(x)$ through the origin; the residue there gives $A(t) = f(x(t))/\big(1 - t\,g'(x(t))\big)$. So A is algebraic, with minimal polynomial obtained by eliminating x, and the recurrence follows from the residual test in the field that polynomial generates. Here the residual is 0, of degree -1, which proves the recurrence for every $n > -1$.

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1 The sequence and the conjecture

OEIS A359643 is "a(n) = Sum_{k=0..n} binomial(n,k) * binomial(4*k,k)". It has offset 0 and begins

$$a(0), \dots, a(7) \;=\; 1, 5, 37, 317, 2885, 27105, 259765, 2523813, \;\dots$$

The entry states, not as a conjecture,

> a(n) = [x^n] (1 + 5*x + 6*x^2 + 4*x^3 + x^4)^n. - *Ilya Gutkovskiy*, Apr 17 2025

that is,

$$a(n) \;=\; [x^n]\, f(x)\, g(x)^n, \qquad f(x) = 1, \qquad g(x) = x^4 + 4x^3 + 6x^2 + 5x + 1. \tag{1}$$

This is the only input taken from the entry, and it was checked against the entry's own DATA before use. In particular *no generating function is assumed*: the entry posts none, and the one used below is derived.

The entry separately carries the following comment:

> Conjecture D-finite with recurrence +81*n*(3*n-1)*(3*n-2)*a(n) +3*(243*n^3-8433*n^2+14984*n-7064)*a(n-1) +2*(-58607*n^3+297306*n^2-491401*n+269124)*a(n-2) +6*(n-2)*(56663*n^2-237722*n+252221)*a(n-3) -3*(n-2)*(n-3)*(111625*n-286402)*a(n-4) +110653*(n-2)*(n-3)*(n-4)*a(n-5)=0. - *R. J. Mathar*, Jan 09 2023

As of the "Last modified" line on the live entry (2025-04-17, revision 25) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

2 A generating function from the extraction

Lemma 1. *Let f, g be rational functions regular at 0 with $g(0) \neq 0$, and let $a(n) = [x^n]f(x)g(x)^n$. Then $A(t) = \sum_{n\geq 0} a(n)t^n$ satisfies*

$$A(t) \;=\; \frac{f\big(x(t)\big)}{1 - t\,g'\big(x(t)\big)},$$

where $x(t)$ is the unique power series with $x(0) = 0$ solving $x = t\,g(x)$. In particular A is algebraic over $\mathbb{Q}(t)$.

Proof. Write the extraction as a contour integral on a circle $|x| = \rho$ small enough that f and g are regular on and inside it:

$$a(n) = \frac{1}{2\pi i}\oint \frac{f(x)g(x)^n}{x^{n+1}}\,dx.$$

For $|t|$ small, $|t\,g(x)/x| < 1$ on the circle, so the geometric series may be summed under the integral:

$$A(t) = \frac{1}{2\pi i}\oint \frac{f(x)}{x}\sum_{n\geq 0}\Big(\frac{t\,g(x)}{x}\Big)^n dx = \frac{1}{2\pi i}\oint \frac{f(x)}{x - t\,g(x)}\,dx.$$

Since $g(0) \neq 0$, the equation $x = t\,g(x)$ has, for small t, exactly one root in the disc, namely the branch $x(t)$ with $x(0) = 0$ furnished by Lagrange inversion; it is a simple zero of $x - t\,g(x)$ because the derivative $1 - t\,g'(x)$ is 1 at $t = 0$. The residue theorem gives the stated value. Both $x(t)$ and hence $A(t)$ are algebraic, being defined by polynomial equations over $\mathbb{Q}(t)$. □

Corollary 2. *A satisfies the polynomial obtained by eliminating x between $x - t\,g(x) = 0$ and $y\big(1 - t\,g'(x)\big) - f(x) = 0$.*

Proof. Both equations hold at $x = x(t)$, $y = A(t)$ by Lemma 1. Eliminating x by a resultant produces a polynomial in t and y vanishing there. □

For A359643 this yields

$$283t^4y^4 - 876t^3y^4 + 930t^2y^4 - 18t^2y^2 - 364ty^4 + 36ty^2 + 8ty + 27y^4 - 18y^2 - 8y - 1 \;=\; 0,$$

and A is the branch of this polynomial whose expansion is the entry's own sequence. The branch is identified by that expansion, not by solving for it in radicals — the polynomial can have degree beyond four, where no such expression exists.

3 The recurrence

Let $\theta = x\,\dfrac{d}{dx}$, acting on x^m as multiplication by m.

Lemma 3. *Let $p_0, \ldots, p_r$ be polynomials and put $b(n) = \sum_i p_i(n)a(n-i)$, with $a(m) = 0$ for $m < 0$. Then $B(x) = \sum_n b(n)x^n = \sum_i x^i(p_i(\theta+i)A)(x)$, so the recurrence $\sum_i p_i(n)a(n-i) = 0$ holds for every $n > d$ if and only if B is a polynomial of degree at most d.*

Proof. Substituting $m = n - i$ gives $\sum_n p_i(n)a(n-i)x^n = x^i(p_i(\theta+i)A)(x)$, and $b(n)$ is the coefficient of x^n in B. □

Since A lies in the algebraic function field generated by the polynomial of Corollary 2, and such a field is closed under θ — differentiating its defining relation expresses the derivative of the generator inside the field — every $p_i(\theta+i)A$ lies there too, and so does B. Computing it exactly gives

$$B(x) \;=\; 0,$$

a polynomial of degree -1.

Theorem 4. *The conjectured recurrence*

$$81n\,(3n-2)\,(3n-1)\;a(n)+3\left(243n^3-8433n^2+14984n-7064\right)\;a(n-1)-2\left(58607n^3-297306n^2+491401n-\right.$$

holds for every $n > -1$.

Proof. Immediate from Lemma 3 and the displayed value of B. □

4 Verification

Every number above is an output of a script written separately from this note, and three independent checks were run.

First, the derived generating function was not assumed to be right. Its series was computed from (1) in two independent ways — once through the residue formula of Lemma 1, by solving $x = t\,g(x)$ as a truncated series, and once directly, by expanding $f\,g^n$ and reading off the coefficient of x^n — and the two agree. The result was then compared term by term with the DATA section of A359643, agreeing on all 13 terms compared.

Second, the recurrence was evaluated directly on the published terms, in exact integer arithmetic and without reference to any generating function: for each n in range the sum $\sum_i p_i(n)a(n-i)$ was formed from the entry's own values and found to vanish, for all 17 values of n from $n = 5$ upward.

Third, the symbolic step is exact throughout. The residual is obtained by rational-function arithmetic in the algebraic function field, not by series truncation, so the identity $B = 0$ is an identity of functions and the theorem holds for all $n > -1$ simultaneously.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org`, 2026. Sequence A359643.

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